

Lesson Exemplar in General Science

Quarter 3

LE

8

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Lesson Exemplar for General Science
Quarter 3: Lesson 8

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LESSON EXEMPLAR

Learning Area	GENERAL SCIENCE	Grade Level	11
Semester	SECOND	Quarter	3

I. OBJECTIVES *(Identifying the Goals)*

Content Standard	The learners learn that climate change affects ecosystems and the organisms living in them.
Performance Standard	The learners use examples to describe the relationships among living systems and their components, as well as how the structure of a part relate to its function. They explain the role of cells in the perpetuation of life and how organ systems work together to maintain good health. They conduct investigation on the effects of climate change in local ecosystems.
Learning Competencies	In this lesson, Grade 11 learners will be able to: describe what is meant by the term “climate change”.

II. REFERENCES and MATERIALS *(Selecting Resources and Material)*

References:

- BBC Earth. “When the World Gets 1°C Hotter | Climate Change: The Facts | BBC Earth.” [www.youtube.com](https://www.youtube.com/watch?v=dIsjcG7hTmo), November 4, 2021. <https://www.youtube.com/watch?v=dIsjcG7hTmo>.
- Environment and Climate Change Canada. “What Is Climate Change?” *YouTube*, March 4, 2021. <https://www.youtube.com/watch?v=QlQ-MEZgRGY>.
- Green Development Sustainable Solutions, Inc. “How Bad Is Climate Change Affecting the Philippines?” *How Bad Is Climate Change Affecting the Philippines?* (blog), May 5, 2023. <https://greendevsolutions.com/climate-change-in-the-philippines/>.
- Ingram, Simon, and Sarah Gibbens. “Photos Show a Climate Change Crisis Unfolding—and Hope for the Future.” *Environment*, 2021. https://www.nationalgeographic.com/environment/article/photos-show-a-climate-change-crises-unfoldingand-hope-for-the-future?utm_source=chatgpt.com.

	Rendon, Jennifer P. "DENR Warns: Extreme Heat due to El Niño Could Ignite Wildfires." Daily Guardian, November 9, 2023. https://dailyguardian.com.ph/denr-warns-extreme-heat-due-to-el-nino-could-ignite-wildfires/. Science ABC. "Climate Change and Global Warming: Explained in Simple Words for Beginners." www.youtube.com, January 16, 2023. https://www.youtube.com/watch?v=G9t__9Tmwv4. Materials Needed: ✓ Computers or tablets with internet access ✓ Paper or digital devices for note-taking and brainstorming ✓ Screenshot or live access to the En-ROADS simulation. Available at https://en-roads.climateinteractive.org/scenario.html?v=25.10.0 ✓ Climate Change Simulator entitled "High Adventure Science: Exploring Climate Change" by The Concord Consortium. Available at https://has.concord.org/global-climate.html	
III. CONTENT <i>(Sequencing Content)</i>	Climate Change	
IV. OBJECTIVES <i>(Setting Clear Objectives and Analyzing the Tasks)</i>	1. Define "climate change". 2. Enumerate the natural and man-made (anthropological) causes of climate change. 3. Identify examples of climate change in the real world. 4. Determine the effects of climate change and suggests ways of mitigating them.	
IV. PROCEDURES <i>(Selecting Strategies, Making Meaningful Content, Delivering Lesson and Assessing Learning)</i>	ANNOTATION	
A. Activating Prior Knowledge	1. Activating Prior Knowledge Option 1: Yes or No	<i>In this lesson, activating prior knowledge is intended to build a bridge between what learners already know and the introduction of new concepts on them about climate change.</i>

	Learners will define climate and climate change through a series of questions generated by them. The teacher will only answer “YES” if the concept is correct and “No” if not. Sample questions that may arise from the learners: -Is climate the same as weather? -Does climate include temperature and rainfall patterns? -Is climate something that can change daily? -Can climate vary between different regions on Earth? -Is climate change only caused by natural processes? -Are greenhouse gases responsible for recent climate change? -Can climate change affect sea levels and extreme weather events? Reflection and Discussion Questions: 1. What was the most surprising thing you learned from the “Yes or No” activity? 2. How did asking your own questions help you better understand the climate change? 3. Did you find it challenging to come up with questions that led to a “Yes” answer? What strategies did you use?	<i>All these activities are designed to engage learners in metacognitive processes such as planning by thinking ahead about what they want to find out or how to approach a question, monitoring by checking if their understanding is correct (via Yes/No answers or feedback) and evaluating by reflecting on what worked, what was confusing, and how they can improve their learning strategies</i> <i>Option 1 Activity lets the learners ask open-ended questions that sparks curiosity, improved motivation, and gave insight into the learners' present understanding. In order to meet learners at their zone of proximal development (ZPD), the teacher could also modify the questions as a follow up technique to elaborate misconceptions among the learners. The idea of this activity is to gradually lead them toward constructing accurate definitions on their own.</i> <i>The questions they may generate help scaffold learning, encouraging critical thinking while allowing the teacher to guide understanding with only “yes” or “no” responses. Once the definitions are inferred, the class could be asked to write their own definitions of "climate" and "climate change" based on what they've unlocked.</i>
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	Option 2: “Climate Change” Alphabet Game The learners will think of words associated with climate change using the letters of the alphabet. Reflection and Discussion Questions: 1. Which word you chose best represents the issue of climate change? Why? 2. Were there any letters that were difficult to match with a climate change-related term? What did that tell you about the topic? 3. If you could replace one word with another more meaningful one, what would it be and why? Note: <i>The learners may answer the activity individually, by pair or by group.</i> Option 3. Build a Definition Together The learners will construct a definition of climate change using the following terms: ✓ long-term ✓ changes ✓ temperature ✓ weather patterns ✓ caused by humans	<i>The “Climate Change” Alphabet Game is a creative and learner-centered activity. Learners must recall or terms related to climate change and creatively match them with letters of the alphabet. This promotes analytical thinking and imaginative use of language. Example, “A – Atmosphere,” “B – Biodiversity,” “C – Carbon Footprint” and so on. Thus, this helps them develop their creative and critical thinking skills.</i> <i>The teacher may also highlight that learners could integrate terms related to but not limited to SHS four curriculum exits, community contexts, etc.</i> <i>The “Build a Definition Together” activity is strongly aligned with the development of 21st century skills, as it encourages learners to think critically, collaborate, and communicate effectively—all while deepening their understanding of a real-world issue on climate change.</i>
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	✓ natural causes Reflection and Discussion Questions: 1. How did each of the given terms help you understand what climate change means? 2. Why is it important to include both human and natural causes in the definition? 3. Why is it important for people to have a clear and accurate understanding of climate change? Note: The teacher may add or change the listed words if he/she preferred a more constructive definition of the term. 2. Establishing the Purpose of the Lesson Option 1: "See, Think, Wonder" This activity uses visual thinking routines to explore meaning. The teacher will show powerful images related to climate change. Examples are images of cracked dry land, melting ice caps, damages brought by typhoons. Then the learners will be asked the following questions: 1. What do you see? 2. What do you think is happening? 3. What might have caused this to happen?	<i>In this activity, the teacher may give emphasis that human activity, particularly the combustion of fossil fuels, deforestation, and industrial pollution, is the primary cause of climate change. The quick changes observed today are caused by human energy and resource consumption, even though natural factors also play a part.</i>
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Sample pictures to be shown:



Option 2: Cause of Climate Change Sorting Game

Sort the following causes of climate change as “Natural” or “Man-Made” (Anthropogenic). Write your answer on the corresponding column.

burning fossil fuels
overuse of fertilizers
transportation emissions
changes in solar radiation

volcanic eruptions
deforestation
open burning of waste
industrial emissions

*This sorting exercise is straightforward yet impactful, as it involves learners in active learning, establishes foundational comprehension, and aligns with the objectives of the **IDF MATATAG** Framework by*

		promoting learner-centered, contextualized, and significant science education.												
	<table><tr><th>NATURAL</th><th>MAN-MADE</th></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>	NATURAL	MAN-MADE											
NATURAL	MAN-MADE													
	Option 3: Climate Change Movie The teacher will show video clips or a short documentary explaining the causes and effects of climate change. After viewing, learners will engage in guided reflection and discussion using prepared questions to deepen their understanding and personal connection to the issue. Reflection and Discussion Questions: 1. What is happening in the video? 2. What causes of climate change were shown? 3. What effects of climate change did you observe? 4. How did the people or animals in the video experience these effects? 5. Have you seen or experienced something similar in your community? 6. What emotions did the video make you feel? Why? 7. What can you do to help reduce the effects of climate change? Suggested videos can be accessed through the following links: ✓ www.youtube.com/watch?v=dIsjcG7hTmo. ✓ www.youtube.com/watch?v=G9t__9Tmwv4. ✓ www.youtube.com/watch?v=QlQ-MEZgRGY.	As learners watch the videos, teachers are encouraged to facilitate reflection by relating the global climate issues shown to local experiences and community realities. Highlight how extreme weather events, rising temperatures, and environmental changes are already impacting lives, especially in the Philippines—such as flooding in low-lying areas, droughts affecting local farmers, or damage caused by typhoons. This contextualization supports the MATATAG curriculum’s emphasis on meaningful, relevant, and resilient learning rooted in Filipino values, critical thinking, and real-life problem-solving. Encourage also the learners to share personal or family experiences, and promote collective responsibility and adaptive action which are the core aspects of the IDF framework.												

	Note: Questions may be answered orally, through pair-sharing, or in writing.	
B. Instituting New Knowledge	3. Developing Mastery Option 1: What happens if we change? Show the class the En-ROADS screen with all sliders at status quo or provide them a copy of the screenshot of En-ROADS interface like the one provided below. Ask learners to observe and take down notes. Guide Questions: 1. What is the projected temperature increase by 2100? 2. Which energy sources dominate (coal, oil, gas, renewables, etc.)? 3. What do the greenhouse gas emissions graph and numbers tell us? Then, show the class how the simulator works. Explain each part and how slider works. Adjust the sliders in the simulator according to learners' suggestions and observe the impact on temperature and emissions. Afterwards, ask them to form small groups. Each group will choose three sliders they would adjust to reduce the temperature increase. Processing Questions: 1. What are the three factors (sliders) you used to reduce the temperature?	<i>The En-ROADS Climate Simulator is an interactive, computer-based tool designed to help users understand how various policies and actions affect global climate change. Developed by Climate Interactive and the MIT Sloan School of Management, En-ROADS models the complex interactions between energy sources, emissions, economic growth, land use, and other factors to show their impact on greenhouse gas emissions and global temperature over time.</i> <i>It has features like:</i> • User-Friendly Sliders - Adjust variables like coal use, renewable energy adoption, carbon pricing, transportation electrification, deforestation, and more. • Real-Time Feedback - See how changes affect greenhouse gas emissions and projected temperature rise by the year 2100. • Educational & Decision-Making Tool - Helps learners, educators, policymakers, and the public explore scenarios and understand trade-offs involved in climate solutions.

2. What was the most effective factor (slider) in reducing the temperature? Why do you think so?

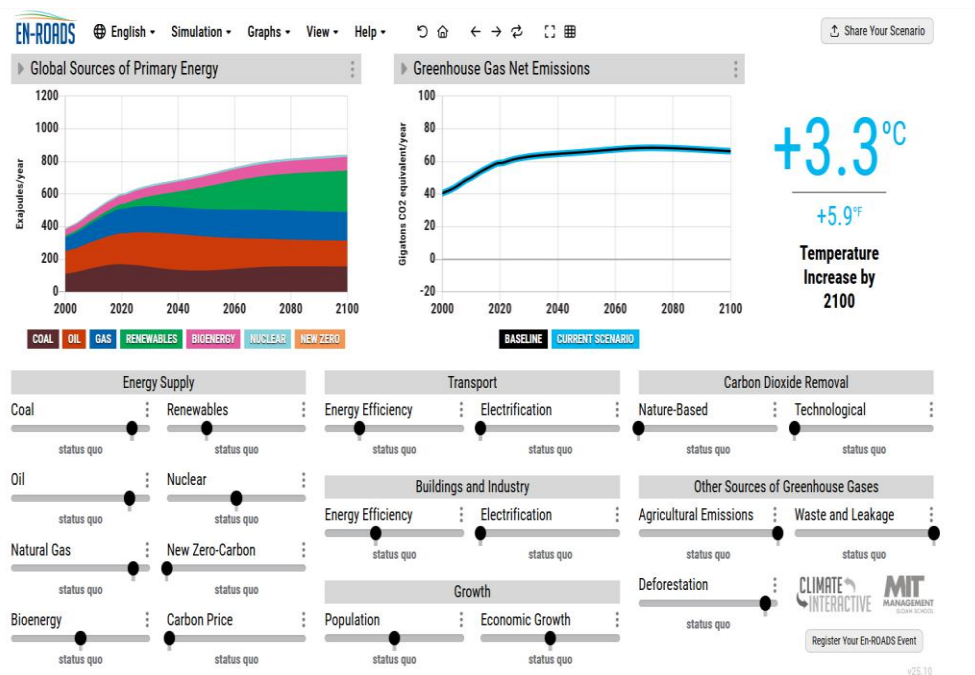
Additional questions:

1. Why do you think the temperature continues to rise even though renewable energy is present?
2. What could be the reasons coal, oil, and natural gas usage remain high?
3. What does a +3.3°C increase mean for the planet? What impacts might we expect?

- **Evidence-Based** - Built on the latest climate science and validated by experts.

This activity applies **hands-on learning** with the En-ROADS simulator to build critical thinking, systems understanding, and responsible action. It helps the learners develop skills in **data analysis, inquiry, and decision-making** like climate experts.

Aligned with the IDF framework, this activity builds prior knowledge, promotes **active exploration, and supports reflection**—key to fostering **scientific literacy** and climate resilience.



Option 2: Exploring Climate Change

Show the learners the climate change simulator entitled “High Adventure Science: Exploring Climate Change” by the Concord Consortium. How its interface looks like is provided in the picture below. Explain how the simulator works by showing the parts and guide them how to manipulate it. Then, let them explore the simulator and ask the following questions:

Guide Questions:

1. What role does CO₂ play in affecting atmospheric temperature?
2. How does an increase in atmospheric temperature influence the amount of water vapor in the air?
3. What happens to temperature when you increased the human emissions by 150% and 200% respectively?
4. Describe how the ice grows or melts as the human emissions is increased.
5. Using the model, what patterns do you observe in how temperature, CO₂, water vapor, and ice interact over time?

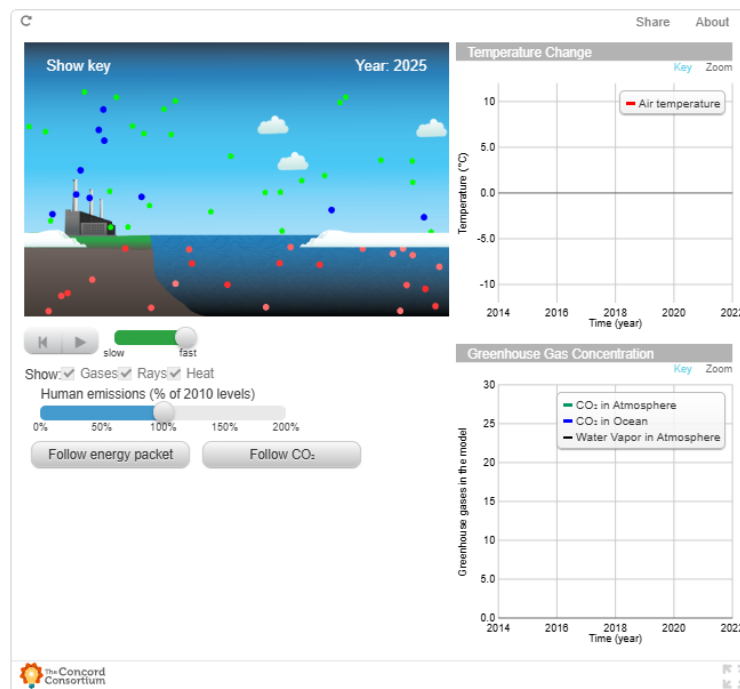
High Adventure Science: Exploring Climate Change is an interactive educational module developed by The Concord Consortium. It focuses on understanding Earth's climate system, particularly how greenhouse gases like carbon dioxide (CO₂) and water vapor interact with temperature. Through the use of the simulator, learners could observe how these gases affect global temperatures and how feedback mechanisms, such as ice melting and water vapor increase, influence climate change over time.

It has features like:

- **Interactive Simulations** - Learners can manipulate variables like CO₂ levels, water vapor, and ice cover to see their effects on global temperature.
- **Real-World Data** - The module incorporates data from ice cores, sediments, and satellites to provide a comprehensive understanding of climate science.
- **Scientific Argumentation** - Learners are encouraged to make claims, explain their reasoning, and express their level of certainty based on the evidence and models provided.

Exploring Climate Change

What is the relationship between CO₂ and water vapor and atmospheric temperature? Discover how the CO₂ level in the ocean and amount of water vapor in the air change as the temperature changes. See how ice growth and ice melting continue once started. Use this model to explore feedback in Earth's climate system, and how they are related to climate change over time.



Note: The learners may answer the activity individually, by pair or by group.

- **Teacher Resources** - Includes a Teacher Edition with background information, classroom tips, and a real-time dashboard to monitor learner's progress.

The teacher should emphasize in using the simulator that carbon dioxide (CO₂) is a naturally occurring greenhouse gas in Earth's atmosphere that helps maintain the planet's warmth by trapping solar heat. However, human activities like burning fossil fuels (coal, oil, natural gas), clearing forests, and various industrial processes add significant extra CO₂ to the atmosphere.

The High Adventure Science: Exploring Climate Change module supports the IDF MATATAG framework by promoting **learner-driven inquiry, real-world problem solving, collaboration, and critical thinking** through interactive climate simulations and data-based exploration.

Option 3: Climate Change Match-Up

The following are the causes and effects of climate change. Match each cause with its corresponding effect from the boxes provided.

Causes of Climate Change

Burning fossil fuels
Deforestation
Melting polar ice caps
Excessive use of electricity
Industrial pollution
Increasing global temperature
Cutting trees without replacing them
Emissions from vehicles and factories

Effects of Climate Change

Increases greenhouse gases, leading to global warming
Reduces Earth's ability to absorb carbon dioxide
Causes rising sea levels and habitat loss
Leads to higher carbon emissions
Contaminates air and contributes to acid rain
Triggers stronger typhoons and weather disturbances
Contributes to desertification and climate imbalance
Adds harmful gases like CO₂ and methane into the air

*The **Climate Change Match-Up** activity builds **conceptual understanding** of climate change through an **interactive and reflective** approach. By connecting causes to real-world effects, learners not only understand the science but also develop a sense of **personal and social responsibility**—a key goal of the **IDF MATATAG framework**.*

	<i>Your answers:</i> Cause 1 _____ Effect 1 _____ Cause 2 _____ Effect 2 _____ Cause 3 _____ Effect 3 _____ Cause 4 _____ Effect 4 _____ Cause 5 _____ Effect 5 _____ Cause 6 _____ Effect 6 _____	
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	Cause 7 _____ Effect 7 _____ Cause 8 _____ Effect 8 _____																
C. Demonstrating Knowledge and Skills	1. Finding Practical Application Option 1: "My Climate Change Action Plan" Learners will identify common sources of greenhouse gas emissions in their daily lives and propose actionable steps to reduce or mitigate these effects at home and in school by complete the chart below: <table border="1"> <thead> <tr> <th>Area of Life</th><th>Common Climate Issue</th><th>Action to Reduce Impact of Climate Change</th></tr> </thead> <tbody> <tr> <td>Electricity use</td><td></td><td></td></tr> <tr> <td>Transportation</td><td></td><td></td></tr> <tr> <td>Waste management</td><td></td><td></td></tr> <tr> <td>School environment</td><td></td><td></td></tr> </tbody> </table> Option 2: Earth Talks Back The teacher will encourage creative engagement among the learners with the concept. They will create an essay with the theme: <i>"Planet Earth explains climate change to humans"</i>	Area of Life	Common Climate Issue	Action to Reduce Impact of Climate Change	Electricity use			Transportation			Waste management			School environment			<i>This activity promotes learner-centered, values-driven education by empowering learners to reflect on their own lifestyle choices and propose practical solutions to reduce the impact of climate change in their immediate environment.</i> <i>In this reflective and creative activity, learners will write an essay from the perspective of Planet Earth, explaining climate change to humans.</i>
Area of Life	Common Climate Issue	Action to Reduce Impact of Climate Change															
Electricity use																	
Transportation																	
Waste management																	
School environment																	

	Option 3: “Climate Change News Gallery” Learners will identify real-world examples of climate change. They will bring a news article or online post (printed or digital) showing a recent event related to climate change (e.g., flood, heat wave, wildfires, etc.). Each learner will give a brief summary of the event, location, and how it's linked to climate change. Determine also whether if they are natural or man-made. Then, propose a possible solution to mitigate its effects. 2. Making Generalization Option 1: “Climate Change” Emoji Choose or draw an emoji that shows how the word "climate change" makes you feel. Guide questions: 1. What is your chosen emoji? 2. Why did you choose such emoji?	<i>This activity engages learners in identifying real-world impacts of climate change by analyzing recent news and proposing relevant solutions, promoting active and contextualized learning. This task encourages inquiry-based learning as learners gather, analyze, and present information. It also supports critical thinking and reflection, allowing learners to examine the causes (natural or man-made) of climate-related events. Learners also engage in constructing knowledge by connecting scientific understanding with real-world issues.</i> <i>This activity invites learners to share their feelings and views about climate change through emojis, helping them build a personal connection and emotional involvement with the subject. By combining self-reflection with scientific learning, it supports the development of emotionally aware and actively engaged learners, aligning with MATATAG’s holistic educational approach.</i>
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Option 2: Wrap it up!

Complete the table below using the concepts you have learned:

Cause of Climate Change	Effects of Climate Change	Ways of Mitigating the Effects of Climate Change
1.		
2.		
3.		
4.		
5.		

Option 3: Climate Change in a Nutshell

The teacher may pose the following questions to the learners:

1. Define climate change in 1 sentence.
2. What are two causes of climate change?
3. Share an example of climate change impacts you've heard or seen.
4. What action can individuals take to fight climate change?

This activity encourages learners to synthesize their understanding of climate change by organizing key concepts about its causes, effects, and mitigation strategies in a structured format. It promotes critical thinking and reinforces knowledge through reflection and categorization.

This activity prompts learners to consolidate and articulate their understanding of climate change in concise terms. By answering targeted questions, learners demonstrate comprehension of key concepts—definition, causes, real-world impacts, and individual mitigation actions.

	3. Evaluating Learning Option 1. “Opinion Line: Agree or Disagree” The teacher will post climate-related statements and learners will reply their stand using metacards showing: ▪ Strongly Agree ▪ Agree ▪ Neutral ▪ Somewhat Disagree ▪ Strongly Disagree Let the learners explain and reflect on their views. Note: <i>The teacher may use the following statements for this activity:</i> 1. Climate change is real and happening now. 2. Even small actions, like saving electricity, can help fight climate change. 3. Cutting down too many trees has no effect on climate change. 4. Human activities are the main cause of climate change. 5. Protecting the environment shows care for future generations. 6. Making small changes in my daily life cannot help fight climate change. 7. People should act now to stop the effects of climate change. 8. Natural disasters like floods and typhoons are becoming worse because of climate change. 9. Schools should teach learners more about climate change and the environment. 10. What I do at home cannot affect the environment.	<i>The "Opinion Line" activity serves as a straightforward yet impactful method to intellectually and emotionally involve learners in the subject of climate change. It integrates effortlessly into the IDF MATATAG framework by promoting reflection, dialogue, and significant learning — all rooted in the real-world significance of climate-related issues.</i> <i>The activity promotes higher-order thinking skills namely: evaluation, reflection, justification. It also fosters a values-integrated learning environment, consistent with the MATATAG framework.</i>
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Option 2: "I Used to Think... Now I Think..."

Complete the statements below:

- "I used to think climate change was..."
- "Now I think climate change is..."

Option 3: Exit Ticket Assessment

1. How do you feel about the topics/concepts discussed? Put a check mark on the corresponding column:

I understood the topics well	I understood some topics	I need more help

2. Can you provide some details supporting your answer?

5. Additional Activities

Option 1: Climate Change Infographic Project

Learners will create an infographic that includes:

- ✓ a clear, simple definition of climate change

These reflective exit activities aim to reinforce **metacognitive thinking** and allow learners to self-assess their evolving understanding of climate change.

In **Option 2: "I Used to Think... Now I Think..."**, learners revisit their prior knowledge and compare it with their current understanding, which encourages **conceptual change** and supports **deep learning**. This aligns with the **Awareness** and **Agency** elements of MATATAG, as learners recognize their own growth and take ownership of their learning.

In **Option 3: Exit Ticket Assessment**, learners evaluate their grasp of the lesson and provide supporting details. This helps teachers **gather formative data**, adjust instruction, and address misconceptions. The activity supports **Goal-Oriented Learning**, as it encourages learners to reflect on whether they have met the lesson objectives.

This activity engages learners in synthesizing and visually organizing essential climate change concepts, fostering deep understanding through creative expression.

	✓ lists or icons illustrating causes (natural and man-made) ✓ images or brief descriptions of real-world examples ✓ effects of climate change displayed with visuals ✓ suggested mitigation actions Option 2: Hot Seat: Ask a Climate Expert One learner will sit in a “hot seat” and will portray as a climate expert. Then the learners will ask some questions related to the mitigating the effects of climate change. They will propose plans, projects or programs that will answer the raised question. Option 3: Climate Change in Our Backyard Learners will investigate and identify how climate change is affecting their local community and propose possible solutions or actions to address its impacts. Guide Questions: 1. What is the local climate issue present in your community? 2. What is/are the possible causes of the climate change issue?	<i>By designing an infographic, learners actively process information—defining climate change, distinguishing natural and human causes, identifying real-world examples, and illustrating effects and mitigation strategies.</i> <i>This interactive activity promotes critical thinking, communication, and collaborative problem-solving by having learners take on the role of a climate expert. The learner in the “hot seat” responds to peer-generated questions related to mitigating climate change, while the class proposes relevant plans, projects, or programs as solutions.</i> <i>This activity allows learners to become critical thinkers, responsible decision-makers, and empowered advocates for climate action. By drawing connections between scientific content and real-life impact—both local and global—they reflect the spirit of a resilient, values-oriented, and future-ready Filipino learner.</i>
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	3. What is/are the effects of the climate change issue in your community? 4. What is/are your proposed solution to the present climate change issue? 5. What is one thing you can do to help address the issue and mitigate its effects?	
V. ASSESSMENT <i>(Assessing Learnings)</i>	Option 1: Multiple Choice <i>Instructions: Choose the letter of the correct answer. Write your answer on the space provided before each number.</i> ____ 1. What is climate change? A. Seasonal rain changes B. The daily change in temperature C. A sudden shift in weather patterns for a day D. A long-term change in Earth's climate patterns ____ 2. Which gas is a major contributor to climate change? A. Oxygen B. Carbon dioxide C. Helium D. Nitrogen ____ 3. Which of the following are natural causes of climate change? A. Volcanic eruptions B. Ocean current shifts C. Solar radiation changes D. All of the above ____ 4. Which of the following are human activities that contribute to climate change? A. Deforestation	

- B. Burning fossil fuels
- C. Industrial emissions
- D. All of the above

____ 5. Which are examples of climate change in the real world?

- A. More frequent heatwaves
- B. Stronger typhoons
- C. Melting glaciers
- D. All of the above

____ 6. Why does deforestation contribute to climate change?

- A. It removes oxygen.
- B. Trees cause pollution.
- C. It cools the Earth too much.
- D. Fewer trees mean less carbon dioxide is absorbed

____ 7. How does the increase in greenhouse gases affect the planet?

- A. It causes Earth to cool.
- B. It speeds up the water cycle.
- C. It reflects sunlight away from Earth.
- D. It traps more heat in the atmosphere.

____ 8. Which action helps reduce the effects of climate change?

- A. Planting more trees
- B. Increasing car usage
- C. Using more plastic products
- D. Burning more coal for energy

____ 9. If a coastal city experiences regular flooding due to sea level rise, what is the most climate-friendly response?

- A. Relocate schools
- B. Create mangrove forests
- C. Build more parking spaces

D. Add more cement to the shoreline

____ 10. Why is it important for individuals to help mitigate climate change?

- A. Climate change only affects big countries
- B. It's not important since it's a natural cycle
- C. Only scientists can solve climate problems
- D. Individual actions collectively make a difference

Option 2: True or False

Instructions: Write T if the statement is True, and F if it is False.

- ____ 1. Climate change only affects the weather and not living things.
- ____ 2. Burning fossil fuels like coal and gasoline contributes to global warming.
- ____ 3. The greenhouse effect is a natural process that keeps Earth warm.
- ____ 4. Cutting down too many trees increases the amount of carbon dioxide in the air.
- ____ 5. Climate change has no effect on the rising of sea levels.
- ____ 6. Recycling and reducing waste are ways to help fight climate change.
- ____ 7. Human activities have nothing to do with climate change.
- ____ 8. Planting more trees can help absorb carbon dioxide.
- ____ 9. Glaciers are melting faster today because of increasing global temperatures.
- ____ 10. Using energy from the sun and wind can help reduce pollution.
- ____ 11. Climate change causes more frequent and stronger natural disasters.
- ____ 12. Animals are losing their habitats because of changes in climate.
- ____ 13. We can help reduce climate change by turning off unused lights and appliances.
- ____ 14. Climate change only affects people in cold countries.
- ____ 15. Everyone, including learners, can help protect the environment.

Option 3: Short Answer

Instructions: Answer the following questions briefly:

- 1. Name one natural and one man-made cause of climate change.
- 2. Give two effects of climate change on the environment.
- 3. What is one thing you can do at home or school to help reduce climate change?

	4. Why is it important to act now to stop climate change?
VI. REFLECTION <i>(Feedback and Continuous Improvement)</i>	<i>This section presents the key highlights and challenges encountered by both teachers and learners during the teaching-learning process throughout the unit. It also includes the adjustments made by the teacher to improve instruction.</i>

APPENDIX

Answer Key:

Assessment:

Option 1: Multiple Choice

Correct Answer	Rationalization
1. D. A long-term change in Earth's climate patterns	Climate change refers to significant changes in global or regional climate patterns over extended periods, not short-term or daily weather variations.
2. B. Carbon dioxide	Carbon dioxide (CO ₂) is one of the primary greenhouse gases emitted by human activities, trapping heat in the atmosphere and driving climate change.
3. D. All of the above	Volcanic eruptions, ocean current shifts, and changes in solar radiation are all natural factors that can influence Earth's climate.
4. D. All of the above	Deforestation, burning fossil fuels, and industrial emissions all increase greenhouse gas levels, contributing to climate change.
5. D. All of the above	Increased heatwaves, stronger typhoons, and melting glaciers are observable impacts linked to climate change.
6. D. Fewer trees mean less carbon dioxide is absorbed	Trees absorb CO ₂ during photosynthesis; cutting them down reduces this natural carbon sink, increasing atmospheric CO ₂ levels.
7. D. It traps more heat in the atmosphere.	Greenhouse gases trap heat radiating from Earth's surface, leading to global warming and climate change.
8. A. Planting more trees	Trees absorb carbon dioxide and help reduce greenhouse gases, mitigating climate change. Increasing car use or burning coal worsens it.
9. B. Create mangrove forests	Mangrove forests act as natural barriers against flooding and erosion and support biodiversity, making them an eco-friendly solution.
10. D. Individual actions collectively make a difference	While climate change is a global issue, small actions by many individuals can collectively reduce emissions and slow climate change.

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Option 2: True or False

Correct Answer	Rationalization
1. False	<i>Climate change affects not only weather patterns but also living organisms, including plants, animals, and humans by altering habitats, food sources, and health conditions.</i>
2. True	<i>Burning fossil fuels releases greenhouse gases, mainly carbon dioxide, which trap heat in the atmosphere and contribute to global warming.</i>
3. True	<i>The greenhouse effect naturally traps heat to keep Earth's temperature suitable for life. The problem arises when human activities increase greenhouse gases excessively.</i>
4. True	<i>Trees absorb carbon dioxide; cutting them down reduces this absorption, causing more CO₂ to remain in the atmosphere.</i>
5. False	<i>Climate change causes glaciers and ice caps to melt and increases ocean temperatures, both contributing to rising sea levels.</i>
6. True	<i>Recycling reduces the need for raw material extraction and energy use, lowering greenhouse gas emissions and waste in landfills.</i>
7. False	<i>Human activities, especially burning fossil fuels and deforestation, are major contributors to recent climate change.</i>
8. True	<i>Trees absorb CO₂ during photosynthesis, reducing greenhouse gases and helping mitigate climate change.</i>
9. True	<i>Rising global temperatures accelerate the melting of glaciers worldwide.</i>
10. True	<i>Solar and wind energy are clean, renewable sources that don't produce greenhouse gas emissions or pollution.</i>
11. True	<i>Climate change increases the intensity and frequency of extreme weather events like hurricanes, floods, and droughts.</i>
12. True	<i>Changing temperatures and weather patterns affect ecosystems, forcing animals to lose or migrate from their natural habitats.</i>
13. True	<i>Reducing electricity use lowers energy demand and carbon emissions from power plants.</i>
14. False	<i>Climate change impacts all regions globally, including hot, temperate, and cold areas.</i>
15. True	<i>Everyone can contribute by adopting sustainable habits and raising awareness about environmental protection.</i>

Option 3. Short Answer

The following are the possible answers for Option 3:

1. Name one natural and one man-made cause of climate change.

Natural cause: Volcanic eruptions

Man-made cause: Burning fossil fuels like coal and gasoline

2. Give two effects of climate change on the environment.

Melting glaciers and rising sea levels

More frequent and stronger natural disasters like typhoons and droughts

3. What is one thing you can do at home or school to help reduce climate change?

Turn off lights and appliances when not in use to save electricity

(Other possible answers: recycle, plant trees, use reusable containers, walk or bike instead of using cars)

4. Why is it important to act now to stop climate change?

Because the longer we wait, the more damage occurs to the planet, affecting future generations and making solutions harder and more expensive.